

Electronic Devices and Circuits Study Notes (EC201)

Academic Insight - Public Sample Revision Summary

Module 1: PN Junction Diodes & Rectifiers

- Shockley diode equation: $I_D = I_S * (\exp(V_D / (\eta * V_T)) - 1)$, $V_T = 26 \text{ mV}$ at 300K
- Small-signal dynamic resistance: $r_d = V_T / I_{DQ}$

Module 2: BJT/MOSFET Amplifiers & Operational Amplifiers

- BJT transconductance: $g_m = I_C / V_T$; MOSFET: $g_m = 2 * I_D / (V_{GS} - V_{th})$
- Op-Amp closed-loop gains: Inverting $A_v = -R_f/R_{in}$, Non-inverting $A_v = 1 + R_f/R_1$
- Note: Sample educational revision sheet provided by Academic Insight.